

Pre-feasibility study for Dates Grading and Packing Center



2017



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1. Executive summary

Palm cultivation in Jordan is relatively recent and has increased significantly over the last ten years. This cultivation is successful in many areas of the Kingdom, especially in the Jordan Valley region. One of the most important palm cultivars grown in Jordan is the Al-Majhoul and Al-Burhi, where these two types represents 85% of total cultivated palm trees. These two types are considered to be one of the most needed varieties regionally and internationally. Therefore, it was proposed to establish a center that aims to provide sorting, washing, grading, packing services for dates in the Jordan Valley, in exchange for a fee of (120 dinars / ton). It is possible to buy the fresh dates from farmers and to carry out grading, packaging and marketing operations, but this requires raising the working capital to carry out the cash purchases from farmers.

The following table shows the main results of the study:

Table 1 Main results of the study

Project name	Dates Grading and Packing Center
Suggested location	Irbid- Jordan valley
Proposed service	Dates Sorting ,grading ,packing , packaging
Total indirect labor (person)	6
Total direct labor (person)	27
Total labor (person)	33
Total investment (thousand JD)	754
Net present value of cash flows (thousand JD)	2,115
Pay back period (years)	4
IRR(%)	%30.9

The idea of the project is to establish a specialized center for steaming, sorting, grading, packing and packaging of dates in Jordan Valley, either by providing this service for farmers or by purchasing dates and taking over the rest of the marketing and other operations in favor of the center.

The project is expected to provide 33 new job opportunities, six indirect staff and twenty seven direct employees. The investment estimated cost of this project is about 758,000 JD. This amount is determined through fixed assets, which include refrigerated warehouses, transmission lines, grading, packaging, forklifts, and a number of other needed equipment, furniture and transport, pre-operating expenses and

initial working capital. The project will be 50% financed through loans and the rest through self financing. If the crop is purchased directly from the farmers, the project becomes more feasible but requires raising the working capital to enable the purchase of the crop by direct cash payment. Despite the high investment cost of the project, it provides many job opportunities commensurate with the size of the investment.

The results of the financial analysis indicates that the net present value of the cash flows of the project is positive, which is an indicator of financial feasibility of the project, and the rate of internal return is much higher than the weighted average cost of capital, which is also an indicator of the feasibility of the project. Other financial indicators indicates that the project is profitable for the investor.

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

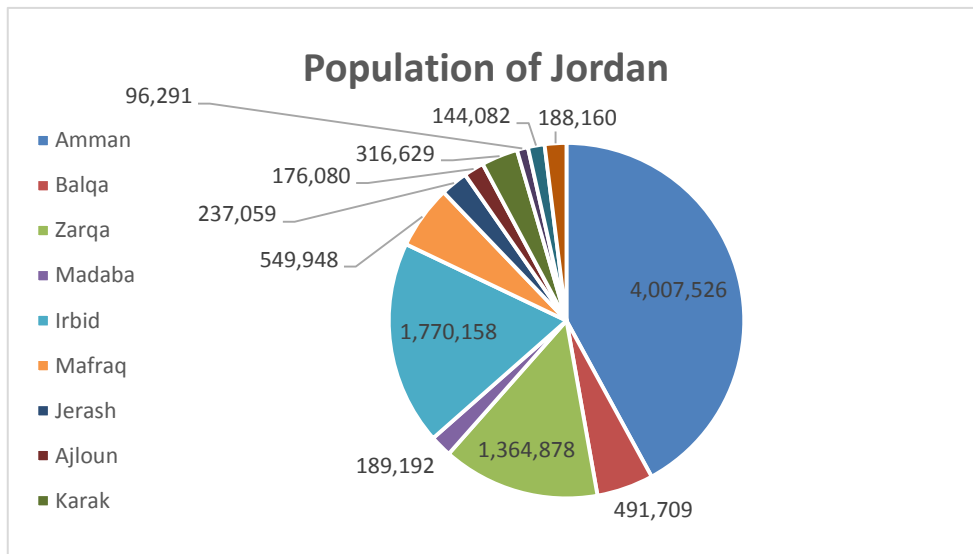
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrqa at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

Department of Statistics (DOS), 2015.

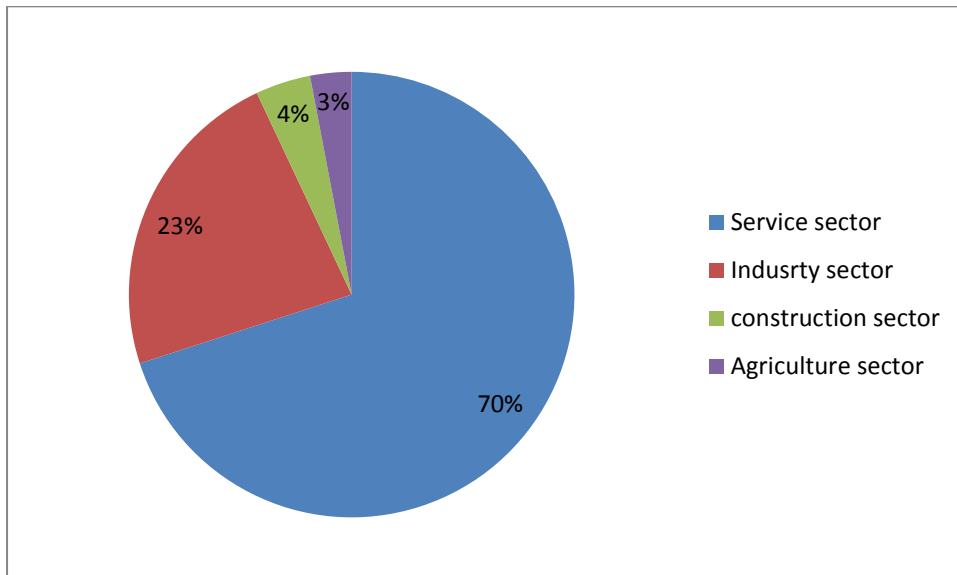
Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has

increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In

addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014.

External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.

- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

The global production of dates is estimated about 7.6 million tons. Egypt occupies the first place in date production, by producing about 1.5 million tons in 2014. Pakistan, Iraq, Tunisia and Saudi Arabia produces about 63% of the world's dates exports. The studies of the Department of Statistics indicate that the number of planted datepalm trees in Jordan increased from 270,000 trees in 2010 to 478 thousand datepalm trees in 2015, with an estimated half a million datepalm trees in Jordan on 27,000 dunums in 2015 (Table 3). Production quantities of dates for the next five years is estimated to be increased to 28 thousand tons, because of the high economic feasibility that drives farmers to go to the cultivation of dates at the expense of vegetables (Figure 3).

Today, Jordan is the 14th largest exporter of dates in the world according to FAO statistics, due to the exportation of a desired type of dates called "Almjhou" or "Almjul", which is one of the finest dates and competing in flavor and quality with other types. The production of the Kingdom from (Almjhou) is exported to the Gulf countries, Europe, Turkey and the United States, "at prices ranging between (7.5-10 dinars per kilogram) depending on size, quality, type and packaging as a result of the transformation of some traditional farmers to grow dates.

Although there are twenty varieties of dates planted In the Kingdom. Almdjol and Berhi occupies about 85% of the total production, and these two types are the first channel for investment in this sector, as about 40% of their production are exported and the rest for the local market, The lands planted with date palm trees exceed 27,000 dunums. Date palm trees are characterized by their ability to withstand harsh environmental conditions. They are the most tolerant trees of water shortage and high temperatures., in addition to the great tolerance to salinity of soil and irrigation water

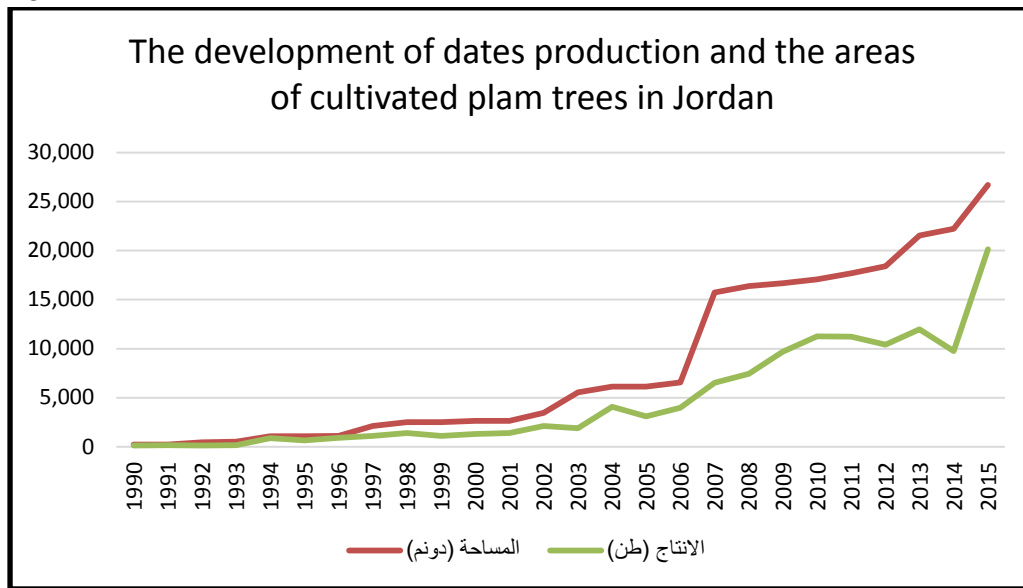
Table 3 Development Number of Palm Trees, production, cultivated area in Jordan during the Period (2010-2015)

Year	Area	Total number of trees	Number of productive trees	production	Average production per tree
2010	17,079	269,528	232,619	11,241	48
2011	17,690	285,620	234,048	11,213	48
2012	18,397	297,145	220,083	10,417	47
2013	21,520	304,735	222,340	11,981	54
2014	22,220	330,032	184,798	9,764	53
2015	26,691	477,833	332,637	20,141	61

In 2015, the estimated number of date palm trees planted in Jordan was about 477,000 trees planted on 27,000 dunums. The dates palm trees are planted in the (Jordan Ghor) along the western edge of the Kingdom, in addition to the Eastern Azraq area where Al-Barhi type is cultivated. The importance of cultivating the dates of the Almjoul and Al-Burahi in Jordan is due to the following reasons:

1. The Jordan Valley provides an ideal climate for the production of high-quality dates especially Almjoul.
2. Jordan has the potential to undertake pre-post and post-harvest operations at competitive costs.
3. Portability of dates storing limits marketing bottlenecks.
4. The high revenues of cubic meters of palm trees compared to conventional crops.
5. Dates are considered to be an export commodity that is globally required.
6. pre-harvest and post-harvest operations depend on employment throughout the year, and women accounts 50% of employment in dates processing.
7. Positive impact on other industries and supporting activities.

Figure3 The Evolution of Production and Cultivated Palm Areas in Jordan (1990-2015)



Project description

The project idea is to establish a center in (Jordan Valley) AL aghouar mainly for sorting, processing and packing of dates and the production of date derivatives of non-marketable varieties with a production capacity estimated at 2400 tons per year of good dates. Such projects are considered agricultural projects and are directly related to packaging, marketing and will help small farmers to market their production in exchange for a fee they pay to do drying operations, storage and sorting and gradualization, packaging and transport.

Such projects are considered as projects that can easily control their size, since most machinery and equipment can be provided locally as required shape and size. Therefore, in order to create the largest number of job opportunities, the suggested production of these factories ranges between 0.5-1 tons per hour/ day. It can be considered pilot projects that need to train the cadres and workers before starting the project. This project could lead to the creation of new employment prospects.

Project objectives

The following points summarize the objectives of the project:

1. The project aims to provide grading and packaging services for farmers in Jordan Valley. There is a demand by small farmers of dates producers as well as the Jordanian Association of dates producers to establish a plant for grading and packing dates in Jordan Valley region. The project aims to achieve high profit for the project owner.
2. Providing distinctive drying, grading, packing, packaging services for farmers.

3. Increasing the exporting capacity of the farmers.
4. Creating job opportunities in the less developed areas, and improving the economic and social conditions of women, since most of the manual processes of sorting and grading are usually done by women.

Proposed product

The Jordanian product of dates is considered as one of the best varieties in the world and is exported to a large number of Arab and foreign countries, and farmers especially small farmers face marketing problems because they do not have grading and filling workshops. Big farmers have gradual workshops for grading and packaging. They do not face marketing problems and they market their products according to export agreements.

The investor Mahmoud Abu Ayash, owner of date palm farm in Karama, indicated that small farmers face marketing problems, especially after the closure of the Syrian border, because they do not have gradual and packing workshops, unlike the major traders who own these workshops, demanding attention to those farmers because they are the main pillar in date palm cultivation,

The project idea is to establish a specialized center of storing, drying, gradualizing, packing and packaging dates that are received from farmers, in exchange for wages, for the production of packed dates in carton containers or transparent plastic bags with different capacities.

In case of the availability of appropriate gradualization and packing of dates, the Jordanian exporters can enter the foreign and European markets prior three to four weeks before the entry of the production of the most international countries that produces dates, such as Egypt, Tunisia and Iraq, so the Jordanian production can get the best international prices for dates in those markets before the entry of competitors, due to climate factors and weather conditions and land suitable for the production of dates.

Market size analysis

Data of foreign trade in the Department of Statistics indicates that the average quantities of imported dates of all kinds (fresh and packaged in containers less than 1 kg and in containers of more than 1 kg) during (2010-2015) was about 11,795 tons imported from KSA and Egypt. Table 4 shows the value and quantity of imports of dates during the period (2010-2015) with an estimated value of (13) million JD. Table 4 shows that there is a deficit in the trade balance for the dates in 2014 amounted about (16) million JD.

Table 4 Imports, exports, re-exports and trade deficit of dates

year	Imports (JD)	Imports (tons)	Import average price (JD/kg)	Exports (JD)	Exports (tons)	Export average price (JD/kg)	Re- exported (JD)	Re- exported (tons)	total exports (JD)	Trade balance (JD)
2010	7,384	8,048	0.92	2,452	1,971	1.24	664	623	3,116	-4,269
2011	8,791	9,635	0.91	3,793	2,711	1.40	579	516	4,372	-4,419
2012	12,540	10,915	1.15	3,390	2,293	1.48	569	411	3,959	-8,581
2013	13,633	11,985	1.14	4,447	3,202	1.39	802	660	5,249	-8,383
2014	22,175	17,535	1.26	5,323	3,561	1.49	796	640	6,120	-16,055
2015	14,797	12,650	1.17	5,628	3,643	1.54	345	292	5,973	-8,825
Average	13,220	11,795	1.09	4,172	2,897	1.43	626	524	4,798	-8,422

Source: Department of Statistics (2017) Foreign Trade Bulletin, http://www.dos.gov.jo/dos_home_a/main/linked-html/external_trade.htm

Most of the palm plantations in Jordan are eaten as ripe dates, and the import comes in the form of dates amounts to 90% of the high-sugar dried varieties. The study of household expenditure for 2013 indicates that the average annual consumption of dates per capita is 0.89 kg per year, which is very low if Net imports were divided minus exports plus grossdomestic production (GDP).

According to the statistics of the Ministry of Agriculture, the annual local consumption of dates ranges between 16 to 18 thousand tons for 2012, which are close to the results shown in (Table 5). The importance of date production in Jordan is that production begins at the end of the production season in the Gulf countries, Which allows the export of dates to these producing countries.

Table 5 Estimation of Domestic Demand for Dates (2011-2017)

Market sizes of dates	2011	2012	2013	2014	2015	2016	2017*
Imports (tons)	9,635	10,915	11,985	17,535	12,650	13,030	13,421
Exports (tons)	2,711	2,293	3,202	3,561	3,643	3,753	3,865
Re-exported (tons)	516	411	660	640	292	301	310
Local production (tons)	11,213	10,417	11,981	9,764	20,141	20,745	21,367
Consumption (tons)	17,620	18,628	20,104	23,097	28,855	29,721	30,612

On the assumption that there will be no increase in the demand for dates (ie, there will be no change in the consumption pattern) and the increase will only be within the annual increase in population growth by 2.2%. There is an additional demand for dates by some cakes factories and muffins and bakeries and also by restaurants, that is estimated about 1000 tons per year as a result of the increase in the number of bakeries and shops that manufacture the Cakes and Maamoul, especially during the feasts. It is difficult to determine the percentage of restaurants' demand due to the lack of accurate data. Thus, the demand for the years 2017-2027 is expected to be as shown (Table 6). The demand for dates is expected to increase from 28.8 thousand tons in 2018 to about 35 Thousand tons in 2027.

Table 6 The expected demand on dates during the period (2018-2027)

Expected amount of demand	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Estimated population (in thousands)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Average consumption per capita (kg/person)	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890
Local consumption in tons	28,855	29,490	30,139	30,802	31,479	32,172	32,880	33,603	34,342	35,098

1.1 Competitors analysis

Although there are no significant problems to market dates for major producers such as the investor D. Abdulla Arar, investor Mahmoud Abu Ayyash, Al-Haq Farms, Kawar, and Samir Rifai. Khattab palm farms and other major producers (Table 7), where they have filling and grading centers, but not more than 10 private centers. Some offers the service in exchange for pay. But there are marketing difficulties facing small producers.

Therefore, it is not expected that the project will face difficulties in achieving the expected sales, as there is a demand and need of small farmers for this service in order to market their products to local markets or export to Arab and European countries.

Table 7 Names of Jordan Dates Association members

Name	Name	Name	Name
Dr. Mohammed Ali Al-Qaoud	Al-Nabulsi farms	Sama farms for dates production	Tariq Hassan / Malik Batayneh
Qais Jafar Farm	Abdulrahman Afaneh Farms	Arar Farms	Fouad Abu Jaber / Al - Anqaa Farms
Qishta farm	Forsan dates farms	Alian Farms	Omar Ershid Abdel Baqi
AL tamayoz company / Ghoul	AL mazaree al moubarkeh company	Al - Sharawi Farms	Hussein Marzouki
AL karam products company	AL haq farms	Tamara farms	Jamil Dasouqi
Abu Ayyash Farms/Al Dafaf Agricultural Products Co.	Soboh farms	AL nawa farms	Ibrahim Rajabi
Majdoul aloudrodn company	AL nakheel farms	Youssef Shehadeh Shamieh	Hamdi Mahmoud Sarah

Al barakeh farms	Iyad almai farms	Haidar Murad & Sons Group	Mansour Al Shehab
Marwan Al - Hamoud Farms	Sha'sha'a farm	Rami Tadros Farms	Haidar Ibrahim Al Moasher
Alwan Mohammed Alawneh	Nizar Alami	Jana dates company	Emad Khaled Ahmed Masad
hammad farms	Mohammed Ali Abu Ayyash	Hamdan farm	Al - Ramla White Farms for Dates and agricultural Products
Wardet al ramel farm	Ali Obeidat	Khatab farm	Ahmad Khalaf Ahmad Abu Dayya
Salah Eddin Amro	Sulaiman Obeidat	Arab International Agricultural Development Company	Advanced Agricultural Investments Company
Wadi Al - Khasib Company	Emas Company	AlKatar farm	Khalifa Farms
Abdel Wali al falahat	Zu al kafal. Ghosheh	Firas Al Adwan Farm	Dr. Zaid Ziad Madi
Refaat Ahmad Salameh	Samara Farm for Dates	AL khair Farms / Umm Qais Agricultural Company	Village of Al majhoul for agricultural land management

Prices analysis and pricing strategies

Dates are marketed to the European markets, especially Turkey, in addition to the Gulf markets and the local market, indicating that the prices of dates in Jordan are moderate and the price of Al-Burhi about 600 fils per kilo, while the Al majhoul is priced according to the weight of the product and its size and ranges

from 2.5 to 8 dinars per kilo. Some top producers of dates provide grading service at very high prices, sometimes reaching 1JD/ kg. Some owners of the grading centers purchase the crop from the farmers and take the harvesting process from the date palm trees (called the farm guarantee). These farmers complain of the low return compared to the return after the processes of packing and packaging and there are some importers are importing dates in large-sized containers and then re-packaged in small containers of polystyrene or plastic weighs about 200 - 500 g or 1 kg.

Through previous studies prepared in favor of the Ministry of Agriculture to establish four centers for the processing and packing of vegetables and fruits in the Kingdom at production centers in Mafraq, Ghor al-Safi, Deir Alla and Madaba. The results indicate that the average cost per kilogram reaches 100-120 dinars per ton, as shown in (Table 8).

Table 8 proposed prices and production rate for the project

Suggested price for provided service	Production rate	Price(JD/ton)
Extra-Jumbo (JD / Ton)	%25	150
First Class (JD / Ton)	%35	125
Second Class (JD / Ton)	%25	110
Third Class (JD / Ton)	%15	70
Expected price (JD / Ton)	%100	119.25

The Jordanian product of dates is considered one of the best varieties in the world and many countries are racing to buy Jordanian products of dates, and Turkish investors recently held negotiations with the Association of date producers to buy the whole crop of Jordan of Al Majhoul type, the area currently planted in the Jordan Valley is more than 17,000 dunums and their production from Al-Burahi and AL Majhoul amount to 16 thousand tons, and their economic return of these two kinds amounts to 28 million dinars. It is expected to increase production in the coming years to 8 thousand tons of AL Majhoul, raising the economic return of this variety to 40 million dinars.

Expected market share

The AL Majhoul is the best kind of dates for many people, which is characterized by its size and taste and its many health benefits. AL Majhoul date is eaten fresh or dried, where it can be stored dry for long periods of time. AL Majhoul grows along the western border of Jordan adjacent to Jordan River extending into Wadi Araba. This area offers an ideal climate.

Most date palm trees cultivated in Jordan are of AL Majhoul type. It is estimated that the number of trees planted by 2015 are about 300 thousand trees producing about 15 thousand tons and the number of farms is about 500 farm according to the Association of date producers in Jordan, and there are many large farms have facilities to manage post-harvest including Sorting, grading, filling and cooling chambers for storing. There are many new established farms did not reach the stage of production. And (Table 9) shows quantities of production and the productive area of dates in Jordan Valley by administrative divisions. The production of is 15,000 tons produced from 250,000 date palm tree in Jordan valleys. The total number of trees in Jordan Valley is 372,000 date palm trees. It is estimated that the production in 2017-2018 will amount to 24,000 tons of dates as a result of new farms that reach the production stage. 3,600 tonnes of dates are currently being exported to EU countries, Japan and Canada, as well as to Saudi Arabia and the UAE after the end of the dates of date production in these two countries.

The project's market share is about 8,000 tons of dates annually in the area of the Jordan Valley, consisting a percentage of 33% of the total dates produced in the Jordan Valley, by proposing the establishment of a center with production capacity (500) kg per hour with an average of 4 tons per day, with a production capacity of 80% of the designed capacity and thus the production capacity is about 8,000 tons per year and the production capacity is gradually increased as a result of increasing the efficiency of the employees to reach 9,300 tons at the end of the life of the project.

Table 9 Productivity of palms in the Jordan Valley according to administrative divisions

Location	Area	Total number of trees	Number of productive trees	Production rate	Production	Expected production
North shuna	2,271	51,033	45,271	63	2,852	3,215
Dier alla	7,585	114,332	88,885	68	6,044	7,775
South shuna	8,046	200,444	107,053	62	6,637	12,428
Gawar as-Safi	731	6,185	5,225	66	345	408
Total Jordan	18,632	371,994	246,434	64	15,879	23,969
Total	26,691	477,833	332,637	61	20,141	28,932

Expected revenues analysis

Dates can be graded throughout the year, where the harvest is stored in large refrigerated warehouses capable of storing large quantities of dates and the materials withdrawn from these stores are on a daily basis to meet the production requirements. Table 10 shows the share of the project from the local market of dates during the period (2018- 2027), and (Table 12) shows the expected sales of the project. For the purposes of the study, that the average price of the service provided was considered (120) JD/ton after carrying out the required operations of storage, cleaning and gradual packaging and the prices are expected to increase by 2% per year.

Table 10 project share from the local market during (2018-2027)

Demand quantity	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population(in thousands)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Average consumption per capita (kg/person)	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890	2.890
Local consumption (tons)	28,855	29,490	30,139	30,802	31,479	32,172	32,880	33,603	34,342	35,098
Expected production quantity (tons)	3,994	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
Market share from the local market (%)	14%	5%	5%	5%	5%	5%	5%	4%	4%	4%

Table 11 production quantity from diffrent dates classes during (2018-2026)

Market share(tons)	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Extra-Jumbo (JD / Ton)	998	1,997	2,037	2,077	2,119	2,161	2,205	2,249	2,294	2,340
First Class (JD / Ton)	1,398	2,796	2,851	2,908	2,967	3,026	3,086	3,148	3,211	3,275
Second Class (JD / Ton)	998	1,997	2,037	2,077	2,119	2,161	2,205	2,249	2,294	2,340
Third Class (JD / Ton)	599	1,198	1,222	1,246	1,271	1,297	1,323	1,349	1,376	1,404
Total sales (tons)	3,994	7,987	8,147	8,310	8,476	8,646	8,819	8,995	9,175	9,358

Table 12 Expected total sales during (2018-2027)

Project revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	3,994	7,987	8,147	8,310	8,476	8,646	8,819	8,995	9,175	9,358
Product price (JD/ton)	119	122	124	127	129	132	134	137	140	143
Total sales (JD)	476,237	971,523	1,010,773	1,051,608	1,094,093	1,138,294	1,184,281	1,232,126	1,281,904	1,333,693

4. Technical analysis

Dates are classified into several varieties depending on the color of the crust and the type of sugars prevailing in the fruit at ripeness and the timing of their early or late ripeness and the proportion of moisture in the fruit, which is the most used and important criterion. Therefore, there are soft dates with a low percentage of sugars, high wetness of over 30%, dates (half dry) and finally dry fruits. Dates pass through five stages to grow until ripeness. Dates are classified during the various stages of ripeness to (Hababuk and then Alkamri where the moisture rate is 85% and the humidity in half dried dates (called Khilal) is 50% and in the soft date is 45% and in ripen dates 20% and less. There are many kinds of dates, In addition to the Al Majhoul and the AL-Berhi there are sukari, Al Khalas, chakra, Khadri, Mtokomi, Hayani, Zaghloul, Halawi, ahmar Talal, Zuhdi and dajlet Noor.

Dates reaches workshops with different quality, to be cleaned and sorted according to quality standards and market need. The sorting is done mainly by naked eye, and by workers and working along a conveyor belt or fixed tables where dates are sorted to different dates according to their color, size and moisture.

The process of gradualization and packing of dates in several stages of productivity, which may vary from one category to another and vary according to the degree of mechanization and investments available, the process can be summarized as follows in (Figure 3): which may vary from one institution to another

1.1.1 Receiving and Storing.

Dates are received and weighed then initial cleaning, and products are cleaned first by removing the rotten and infected fruits to ensure that a product that meets the health standards. The dates are stored in large refrigerators with a temperature below 0 ° C. The refrigerated storage system aims at preserving the product and preventing any vital danger.

1.1.2 Sterilization

Dates are sterilized into sterilization chambers to sterilize dates and this process based on air discharge by a suction pump with the capacity of 60 horsepower. Dates are sterilized using health-tolerant gases, such as aluminium phosphide gas, CO₂ and aluminium phosphide pills. The capacity of the sterilization chamber can reach to 25 tons.

1.1.3 Drying

Solar energy is usually used in the drying process by spreading the dates into wooden trays covered with gauze or nets to prevent insects and dust from entering for 7-8 days depending on the intensity of the solar brightness.

1.1.4 Dates sorting

Sorting is done in the factory depending on customers orders, where sorting is done depending on size, type, color.

1.1.5 Scaling and grading

Dates are sorted depending on sizes required by the customer, that is done by using dates grading machine, which works automatically, there should be more than one grading machine in the plant with a capacity of productive machine 1 ton/hour

1.1.6 Polishing and moisturizing

The dates are cleaned, moistened and sorted according to their types based on the customer's demand For example, AL sokari dates are sorted into, mashed, wet peel and the bad fruits are extracted. The sorting process is carried out on automatic conveyors with the participation of the labor force. The factory must have a number of automatic conveyerbelts to sort dates; and the production capacity varies according to the number of workers involved in the sorting process.

1.1.7 Dates

Packing process is done by using machines or manually, and during the packing process dates pass through two stages:

- Dates washing: washing process is done automatically using sprayers, that distribute water in the form of spray , so not to affect the external crust of dates.
- Furnace: after washing process , dates enters to furnaces to prepare them for packing process.

1.1.8 Dates Packaging

There are two ways of packaging: flexible bottom and hard bottom.

- The flexible bottom: In which the dates are pressed after adding a certain percentage of water and the process of packing and packaging are done automatically those packs are

known as vacuum packages, that are available in different weights, such as 350 g, 500 g, 1000 g or as customer demands.

- The hard bottom: Where the dates are filled and packaged automatically using latest thermoforming machines for packing dates in plastic and carton containers of various sizes and shapes according to customer demand, in addition a sticker can be placed on the carton according to the customer request.

1.1.9 Cold storing

Dates are stored in cold warehouses until it's marketed, and transported as per determined conditions and specifications

1.2 Jordan Standard Specification

To ensure the export of dates to the Arab and European markets, please have a look at, Jordanian, Gulf and European standard specifications, which are almost similar

Standard No: Jordanian Standard (JSS 549: 2004):

-replaced the standard No. 549 of 1988

Title: Vegetables, fruits and their products - Dates

Abstract: These specifications are concerned with the requirements that must be met in whole dates, with or without pulp, commercially processed and prepared for direct human consumption, and not for the products of other dates such as cut or mashed or used for industrial or stuffed purposes

Status: Jordanian Standard (SAS)

Price (20) JD

Legislative: Mandatory

Keywords: Fruit - Vegetables - Dates ICS

-Fruits and derived products

Standard No. 549: 2013

Title: Vegetables, fruits and their products dates

Abstract: This Jordanian standard is concerned with the requirements that must be met in whole dates, with or without pulp, commercially processed and prepared for direct human consumption, and not for other date products such as dates cut or mashed or used for industrial or stuffed purposes.

Status: N (Final Project)

Price : 20 JD

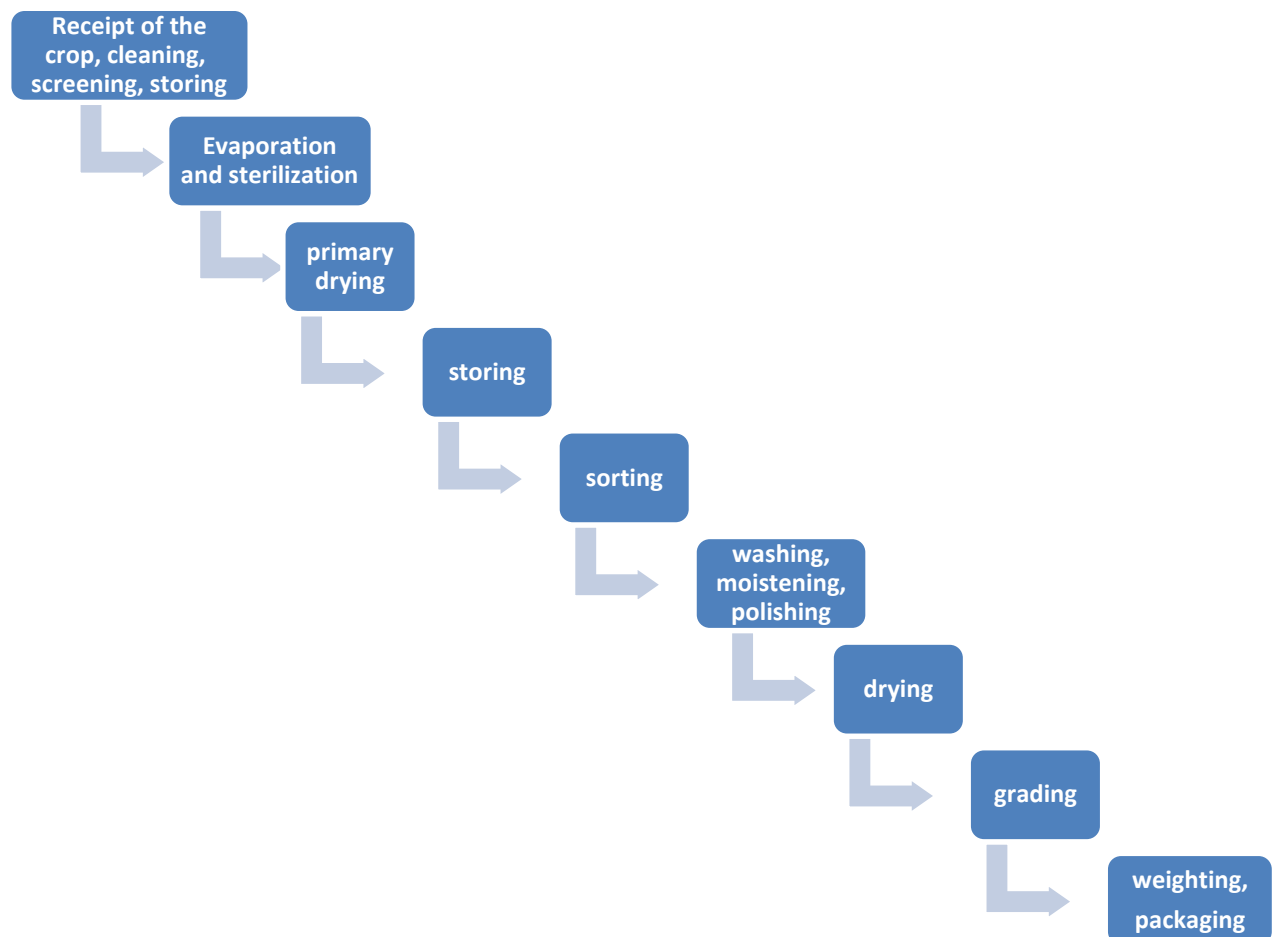
Legislative: Optional standard specifications

Keywords: fruits, health requirements, standard requirements, packaging, pass, statement, vegetables

Production steps

The following figure shows the production processes:

Figure4 production steps



Required raw materials

Raw materials include water, electricity, sterilization chemicals, fumigation, and cleaning materials as well as plastic and carton packaging. Table (13) shows the cost of raw materials.

Table 13 Quantities and cost of the annual required raw materials

Production requirments	Unit cost	Needed quantity	Total value
Water	1.5	2,689	4,034
Electricity	0.1	80,671	8,067
Chemicals for sterilization and evaporation	2	4,034	8,067
Cleaning materials	0.5	40,335	20,168
Packing materials	0.1	2,016,768	201,677
Total			242,012

Required human resources

The project requires agricultural engineer specialized in plant production or food manufacturing, in addition to agricultural processing technician, marketing officer, quality officer, accountant, warehouse officer, forman, and workers for grading and filling, and drivers. The table (14) shows required human resources and their estimated wages :

Table 14 details of the required human resources

Job title	Number	Monthly salary (JD)	Total monthly salaries	Total annual salaries
Indirect employees				
Project Manager / Agricultural Engineer	1	1,000	1,000	16,000
Technical food production worker	1	750	750	12,000
Accountant and warehouseguard	1	600	600	9,600
Quality auditor	1	500	500	8,000
Marketing officer	1	600	600	9,600
Guard	1	300	300	4,800
Total indirect employees	6		3,750	60,000
Direct employees				
Maintenance Technician	1	400	400	4,800
supervisor	1	400	400	4,800
Sorting,packaging workers	20	350	7,000	84,000

Job title	Number	Monthly salary (JD)	Total monthly salaries	Total annual salaries
Driver	3	400	1,200	14,400
Cleaner	2	250	500	6,000
Total direct salaries	27		9,500	114,000
Total salaries	33		13,250	174,000

The administrative costs include the salaries of the project's executive director, sales staff, food processing technician and guarding. The cost associated with the administrative costs is about (60,000)JD per year. The direct labor salaries will be increased due to the accumulating experience to reach 2% per year. In addition ,workers' health insurance and Social Security contribution and trasportainon allowance have been added. The annual salaries of all employees were calculated for 16 months instead of 12 months to cover these costs in order to estimate the total annual labor wages including indirect labor wages as shown in Table 15 .

Table 15 Salaries and wages of direct and indirect labor

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total indirect salaries	60,000	61,800	63,654	65,564	67,531	69,556	71,643	73,792	76,006	78,286
Total direct salaries	114,000	156,560	161,257	166,095	171,077	176,210	181,496	186,941	192,549	198,326
Total	174,000	218,360	224,911	231,658	238,608	245,766	253,139	260,733	268,555	276,612

The following shows the job description for the required human resources:

- Agricultural engineer: Preparation of the daily production plan and the quantities required for grading and packing and time scheduling of production process, determining the number of workers and machines used in each patch, development of appropriate technical solutions to problems related to the production process or the quality control of the products. Identify the raw materials involved in the production process such as temperature, control of stored materials, follow up the process of receipt of dates, weight, and check their varieties and quality control and validity for consumption, matching specifications and keeping them in refrigerators and cold storehouse. Supervising the process of drying, storage, sorting, packing and packaging. Supervising the manufacturing process and water control, evaluating the performance of staff and labors, identifying their training needs, training and developing their skills. Mobilization of business models. Applying occupational safety and health procedures and instructions.
- Food processing technician: receipt of dates, drying weighting, and check quality to match specifications. Sort the dates first by size, type and / or classification level, isolate the damaged pieces, remove the fermented, spoiled parts as requested, and place them in special clean containers. Processing washing machines and disinfection, check machine readiness to work and clean it, and adjust the temperature and time and the proportion of disinfection, sterilizer and operation. Control the washing and sterilizing machines and process. Take samples of finished dates, and check their cleanliness and specifications. Fill the product with appropriate packaging and weights, and store it in suitable storage conditions in terms of temperature and humidity. Cleaning of washing and disinfection and sterilizing machines. Mobilization of business models. Application procedures and instructions of the occupational safety and health.
- Marketing and Supply Representative: studying market trends, prepare marketing plans and programs. Examine the customers' needs on an ongoing basis, monitor the market activity of the different commodities, follow up on securing the customers' requests on time, ensure that the value of the goods sold is met, and set the sales promotion policy. Review the market prices of the products, issue the necessary decisions to adjust them according to prevailing prices, develop a plan for publicizing the media and supervise its implementation, and evaluate its effectiveness. Develop technical and administrative supervision procedures for staff and labors, evaluate their performance and

achievements, and propose programs to increase their knowledge and develop their abilities. Develop procedures and ensure safety and occupational health and public facilities.

- Accountant/Storekeeper: Receipt and deposit of cash in the bank, organization of Journal, Ledger accounts, preparation of the balance of the audit, list of debts and losses, preparation of the balance sheet and tables of goods at the end of the period and the list of property rights; control of records for production and sales, revenues and expenses; Regulatory procedures and instructions for production control, preparation for the receipt of products, inspection and conformity of quantities, types and specifications with the purchase order. Filling out the sales process, selling the product, accounting the customers, editing the invoice, inventory of the contents of the stores, identifying the needs of the company from the production requirements, preparing lists of shortages, securing and storing them. Regulation of sales records and financial records, mobilization of business models and application of occupational and public safety and health procedures and instructions.
- Quality Supervisor and Laboratory: Supervising the receipt of dates, taking a sample to inspect the chemical and biological pollution, taking a sample of each product type, and checking it to ensure that the product meets the required specifications. Follow up the supervision of the cleaning of equipment and tools and manufacturing, and determine the appropriate temperature and humidity for the storage of raw materials and food products and the validity period, and match the results of laboratory tests of the samples with quality specifications, filling the work models. Application procedures and instructions of the occupational safety and health.
- Production workers: Working under the supervision of agricultural plantations, processing and operating dairy processing machinery and equipment, manufacturing dairy products, inspecting product conformity to the required specifications, cleaning equipment, tools, and processing of packaging tools, receipt and packaging of products and packaging and storage until marketing. Processing of packaging tools and accessories of packaging. Placing the packaging on the production line, sorting the product, and classifying it according to the required specifications. Fill the product in the custom packaging, and pack it according to instructions. Collection and packaging of containers, and storage in warehouses or warehouses until marketing.
- grading and sorting laborers: Working under the supervision of a food processing technician and receiving dates and matching specifications, sorting and processing, and

processing the tools of packaging, and receiving the fruits (date) from the field, and sorting and sorting of fruits, packaging and packaging, and storage until marketing, Loading and unloading of boxes manually or automatically from and to the truck, loading and unloading boxes from warehouses and transport trucks, arranging them in designated places and leaving large corridors in the order of boxes. Applying occupational safety and health procedures and instructions.

- **Repair and Maintenance Technician:** Works independently or under the supervision of the agricultural engineer, maintains machinery and equipment, prepares suitable operating programs, prepares cleaning and disinfecting programs for machines and equipment, and determines the materials allowed for cleaning and disinfection operations. Training of workers on how to operate and close them, and maintenance of stores and storage rooms for production supplies and ready-made goods, know the correct ways to clean machines, and know the privacy of each machine separately. Inspect materials, equipment and tools for cleaning. knowledge of each machine, how to use and repair it.. Inspect the equipment and tools. performing tools cleaning, check machines for cleaning in coordination with machine operators, clean machines according to manufacturer's instructions, collect cleaning waste in designated areas, and discharge liquid waste according to instructions. Maintain the cleanliness of the work site, and apply the procedures and instructions of occupational and public safety and health.

Location analysis

Depending on these data, it is suggested to establish a center for grading and packing dates in Jordan central valley , the location was chosen due to the existence of infrastructure of water,electricity, availability of raw materials, and the prevailing production system

Production processes description

The following shows manufacturing process description for dates after the harvest.

1. Harvest the dates in the field:

When the trees are high it is recommended to use stairs or hydraulic cranes to reach and collect the dates and avoid using palm fronds for workers as method for collecting due to weakening the of trees.

2. Transportation from the field

The dates are placed in perforated plastic boxes after being collected, the boxes aren't so deep in order not to squeeze the dates and lose their natural appearance. In the case of Al-Burhi after the separation of the clusters, it is advisable to transfer them without pressure of the fruit as a result of placing the clusters on top of each other, instead it should be transported hanging in a specific vehicle to the place where cleaning and then sorting, polishing, packing and cooling processes are done.

3. Initial sorting

The fruits are sorted excluding the damaged, infected and flattened and the healthy fruits are placed in perforated plastic boxes and then sent for drying.

4. Drying

- Drying will be carried out either in greenhouses with fans for wet air suction or on wooden surfaces shaded with a plastic mesh with holes or white gauze to prevent entry of insects. The drying is done by the indirect heat of the sun or in rooms equipped with a heat source. At a temperature of about 30 ° C.
- Fruits remain about 2 to 5 days to reduce the moisture content in them, which is usually about 50% at harvest until it reaches about 25-30% for Al-Majhoul type.
- For other varieties until the moisture reaches 22 - 18 ° C (achieving the stage of dates)

5. Sterilization:

- The process is carried out using aluminum phosphide gas that is universally accepted, and there are suggestions for using the heat for the dates to reach a temperature of 50-55 ° C where the insects and their eggs are killed. But the heat method has been questioned about the possibility that high temperature may negatively affect the appearance and taste of fruits.

- When using aluminum phosphide gas, the perforated boxes are placed one on top of the other with no more than 10 frames inside tightly locked chambers to prevent toxic gas leaks and using one pill per cubic meter and left for 2 - 3 days (the higher the temperature, the shorter the sterilization period). Instructions are usually available with sterilizer pills. The air is drawn by fans for gas disposal for one to two days, after which the dates are ready for cleaning, sorting, packaging, and for a pest free storage.

6. Polishing and moisturizing:

- dates are transferred after being stored to the packing rooms, which are clean rooms and space suitable to allow easy work before filling. Dates are cleaned manually or mechanically. In the case of mass production mechanization of washing and drying and sorting and packing are usually done, but in the case of small farms they can be done manually.
- The cleaning process begins with a partially moistened small brush and the polishing process is then performed
- Physical and chemical methods are used for polishing process, such as high temperature, steam humidification, or by using of liquid sugar, molasses, starch derivatives, cellulose derivatives or commercial pectin.

7. Grading

This process is followed by the removal of deformed, and swollen peels, or that got undesired color from the sun. After that the sorting by size will be done. In Al-Majhoul grading sorting shall be according to the following sizes:

- Jumbo size - the weight of the date is about 30-35 grams (about 30-30 dates per kilo)
- Large size: the weight of the date is about 25 grams (about 40 dates per kilo).
- Average size: the weight of the date is about 20 grams (about 50 dates per kilo).
- Small size: the weight of the date is about 17 grams (about 60 dates per kilo).

8. packing

After sorting, the fruits are packaged in cartons with different weights 1/4, 1/2, 2 kg and 5 kg, or packed in cardboard, plastic or cork containers. The shape, cleanliness and beauty of the packaging is a factor for attracting the consumer and thus facilitation of marketing process and increasing the revenues.

9. storing

Al-Barhi and some other varieties are harvested in the semi ripen phase when the dates are yellow without a green color and the content of the tannin is low. Most of dates are harvested in full ripening (semi ripe dates and dates) when the fruit content of sugar increases then the content of moisture and the substance of tannin are less, then it becomes more softer and fresher. The good qualities of dates are the size, texture, cleanness and the color of the dates. and free of defects such as sun blight, insect damage, sugar transfer to fruit surface, bloating of crust and mildew.

As for sweetness, sucrose is the predominant sugar in some types, Single sugars predominate in other types, usually total sugar is about 50% of fresh dates or about 75% of dry dates. After filling the dates (the moisture is about 18 C) they will be transferred to the storge which is a clean room with polished grounds and walls have a door and double windows (ordinary door and wire door) to prevent the entry of insects and the room is provided with air conditioning so the temperature don't exceed 20 degrees Celsius, and every ton needs a space of 3 cubic meters in the storage room. Before storage, the store should be cleaned and sprayed with an insecticide. Dates are constantly inspected and evaporated with aluminum phosphide if any pest infection is shown.

A) conservation temperature and humidity:

- Freezing - Temperature 18 - 22 Celsius below zero and the temperature of 5-15 C below zero is the degree of freezing dates -
- Cryopreservation: Temperature of 4 - 6 degrees Celsius below zero. All items can be saved by refrigeration The preferred types of preservation are Al-Khalas. Al-Burhi, Nibut Saif.

In general, most dates can be stored at a temperature of 0 ° C from 6 to 12 months depending on the type of dates. Semi-humid dates such as the Dijla and Halawi can be stored longer than wet types such as Al-Majhoul and Al-burhi, and for longer storage period, the temperature of minus 18 ° C is used.

- **Al-Burahi storing**

Al-Burahi is stored at semi-ripe satge for the purpose of prolonging the marketing period at a temperature of zero and humidity of 70-75% for temporarily stoping the period of moisturizing and the available experiments in this field are limited and indicate that it can be stored from one and a half to two months if marketing takes from one to two weeks from taking it out of the store as the process of moisturizing begins when it leaves the store. It is also recommended that AL-burahi be harvested at a pre-mature stage and sterilized before starting storage - preferably stored in clusters as this may help the delayof moisturization a few days.

- **Al-Majhoul storing**

It can be stored at a temperature around zero and humidity of 70% -75% for 2 - 3 months. It can be stored for one year at 18 ° C below zero without moisture control.

- **Notes**

Ethylene: At 20 ° C, dates are produced at less than 0.1 mg / kg per hour. Unripe dates and dates do not produce this material at this temperature. Ripe dates are not affected by ethylene but can absorb other crops odors. Therefore, it is not recommended to store dates with garlic, onions, potatoes or other crops that have strong odors.

Controlled Environment: Dates are stored in a nitrogen, free-oxygen environment that reduces the color change of the dates and prevents insect infection.

Physiological and physical changes: The change in color to dark brown is due to the enzymes and others components. This change in color increases with increasing humidity and heat, and this change can be stopped by reducing oxygen levels in the storage room.

Acidity: Fermentation (by yeast) Acidity can occur if humidity exceeds 25 ° C in the storage rooms, usually it's less than 20 ° C in the stored dates. **Sugar spots:** When crystallization of sugar under the crust or in the fruit body of humidity types, like Al-Majhoul, affects the appearance of the fruit and changes its texture, but it does not affect the taste of dates. Storing at temperatures above -18 ° C reduces this phenomenon to a great extent, which is common in types that contains sugar Fructose and Glucose.

Pathological diseases: Yeast can cause some negative effects as rotten and bacteria, for instance yeasts, can attack dates and cause the appearance of alcohol (date fermentation), and some bacteria such as cyto bacter can convert alcohol into acetic acid. Bacteria can grow on dates with relatively high humidity, especially if harvested after rain or high humidity in the atmosphere To avoid such negative aspects it is advised to:

- Dry moisture indates to 20 percent or less and this reduces the occurrence of fermentation or mildew.

- Control the appropriate temperature and humidity during the storage stages.

Avoid temperature fluctuations in storage that cause moisture deposition on dates, this can help the rot process by fungi and yeasts.

- Insects can attacke dates store and therefore the stores should be sterilized by gas and then the dates should be packed in sealed boxes against insect entry.

- Storage at temperatures below 13 ° C will prevent the attack of insects as well as stop their reproduction. Storage at a temperature of 5 ° C or less will fight insect attack totally .

5. Technical requirements analysis

Land and construction works

The project needs an area of 2000m², 1500m² of which is for the building and production area, and the rest of the area is for services and other facilities, the estimated price of 1m² is 25JD/m². Thus the cost of the land is about 50,000 JD

The project needs the following buildings and facilities: management office, production hall, refrigerated warehouses, laboratory rooms. The cost includes construction work required for the project and the design of the building and the necessary plans for the project and the drilling and paving, squares and garages. (Table 16) as follows:

Table 16 required construction works

Construction works	Cost(JD/m ²)	Area(m ²)	Total (JD)
Administrative offices	150	150	22,500
Hangars, buildings	150	500	75,000
Refrigerated warehouses	150	500	75,000
Open spaces and garages	10	200	2,000
Cafeteria and workers' restrooms	100	150	15,000
Fence and gate	10	500	5,000
Other works	-	-	18,950
Total cost			213,450

Machinery and equipments

The machines and equipment for the process of grading and packing of dates include many production lines, refrigerated warehouses, cars and other equipment. The following is a description of the production lines,

- Technical specifications of production lines:
- Maximum capacity: 500 kg/hour
- machine structure made of stainless steel that is suitable for food industries .

The line consists of the following equipment:

1. Dates Washing and drying line

Specifications of line machines:

Dates Grading and Packing Center

1. Conveyor belt:

- Dimensions: length 2500 mm, width 800 mm.
- The conveyor belt is equipped with a tank to empty the dates with a capacity of 100 kg.
- The motor is equipped with an electric motor with a capacity of 0.55 kW.

2. Vibretor (sifting) for impurities and chopsticks:

- Dimensions: length 2000 mm, width 500 mm
- Vibrator with 0.55 kW electric motor

3. Conveyor belt for manual control:

- Dimensions: Length 3500 mm, width 1180 mm
- The belt is equipped with barriers to prevent dates falling
- The motor is equipped with a 1.5 kW motor

4. Dates washing machine:

- Specification: Length 3000 mm, width 600 mm
- The machine is equipped with a pump for washing with water spray heads
- The machine is equipped with an air turbine at the first drying stage after washing

5. Drying Oven:

- A. Dimensions: Length 7500 mm, width 600 mm.
- B. The oven is equipped with electric heaters, electric capacity of the heaters: 36 kW
- C. The oven is equipped with a 1.1 kW engine.
- D. Drying rate in oven: 90% ~ 100%

2. Sort date line

Specifications of line components:

1. Conveyor belt:

- Dimensions: length 2500 mm, width 400 mm
- The belt is equipped with a tank to empty the dates with a capacity of 100 kg
- The motor is equipped with an electric motor with a capacity of 0.55 kW

2. Dates sorting machine:

- The machine sorts the dates into four measurements
- Dimensions: length 3000 mm, width 650 mm
- The machine is equipped with an electric motor with a capacity of 0.75 kW

3. hydraulic date piston

The date piston is a semi-automatic hydraulic piston. The piston presses the dates into the mold, where the dates are produced in the form of the mold.

- The piston is equipped with a 2KW hydraulic set
- Standard piston equipped with one piston only

4. Thermal packing machine for forming and packing dates with air drag

The machine will thermally form the covering plastic film, the dates are filled, and then welding the lid, then air vacuum from the package. The machine operates with compressed air system and mechanical system

Table (17) shows the project needs of machinery and equipment needed for the project.

Table 17 required machinery and equipment

Required machinery and equipment	Qty	Unit cost (JD)	Total cost (JD)
Initial cleaning line	2	10000	20,000
Sorting and grading lines	4	50000	200,000
Refrigerated warehouses	8	15000	120,000
Washing and humidifying unit	1	5000	5,000
Sterilization unit	2	5000	10,000
Belt conveyer	4	5000	20,000
Packing and pressing machine	1	5000	5,000
Drying unit	2	10000	20,000
Tables and chairs	40	500	20,000
Control unit and laboratory	1	5000	5,000
Water cooling unit	1	3000	3,000
Packing and air-drawing machine	2	1000	2,000
Electronic balance	4	250	1,000
Forklift	2	1500	3,000
Field filling boxes, capacity of 5, 10, 20 kg	2.5	1000	2,500
Generator, air conditioners	1	10000	10,000
Other equipments	1	-	44,650
Total (JD)			491,150

Laboratory machines and equipments include sensitive balances, water bath, centrifuge, pH meter, glassware, chemicals and other necessary tools to check for bacterial contamination and quality of production .

Furniture

The project requires simple furniture for management and permanent employment, that is estimated about 4000 JD, which is simple tables and offices, cabinets and laboratories' furniture and workers' resthouse.

Vehicles and transportation

The project needs refrigerated transport vehicles to transport the product to the market, transporting raw materials and workers, and for a medium transport vehicle for marketing the production to the targeted markets as shown in Table 18.

Table 18 required vehicles

Required vehicles	Qty	Unit cost (JD)
Medium vehicles	1	15,000
Refrigerated vehicle	1	25,000
Total cost	3	40,000

Essential Services

Electricity, water and transportation services are available in most parts of the Kingdom and are assumed to be available in the project site. The needs of services for the project are estimated to be about 2% of the total annual revenues as shown in Table 19.

Table 19 Project needs of support services

Expense Hypotheses	
Maintenance expenses from revenues	0.1%
Consumables expenses from revenues	0.1%
Insurance expenses from the assets	0.0%
Marketing expenses of revenue	1.0%

Legal requirements and procedures

The dates grading activity requires licenses from the Ministry of Agriculture. The project seeks to produce a prepared foodstuff for local markets and exporting to Arab and foreign markets. Therefore, before starting the investment, a number of legal and administrative conditions must be met. These include the registration of a legal establishment, obtaining of necessary licenses and approvals, etc. Delivery of required documents and payment of the initial administrative costs associated with the investment.

1. Legal registration whether it is an individual institution or a company (solidarity, simple recommendation, private contribution, limited liability company, etc.)
2. Registration document of land or lease contract for the land or buildings on which the project is to be constructed on
3. Registration of the trade name of the company .
4. Registration at the Chamber of Industry / Commerce. Every establishment engaged in industrial or commercial activity must be registered in the Chamber of Industry / Commerce
5. Professional licensing - is an essential prerequisite for legal entities to be able to carry out business activities.
6. Registration at the General Organization for Social Security - Every Jordanian legal establishment under the provisions of Jordanian law is committed for registration in social security
7. The approval of the Ministry of Agriculture is required in accordance with the provisions of No. (34/34) for the year 2016 issued in the Official Gazette. These instructions relate to the licenses of the centers for the gradualization and packing of vegetables and fruit produced under Article 58 of the Agriculture Law No. 13 of 2015 and its amendments. Which set licensing conditions for this work
8. The approval of the Food and Drug Administration for dates in accordance with the Jordanian standard specifications after studying its components and the card of its statement and its use. The Food and Drug Administration shall ensure the provision of health requirements to ensure the validity of the product for human consumption.

6. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 20 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 21 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 22 working time assumption

Working time	
No. of weeks	52
No. of days per week	6
No. of days annually	312
No. of working hours	8

Table 23 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 24 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	0.00%
Maintenance Expenses from Total Revenues	2.00%
Depreciation Expenses from Total Revenues	2.00%
Insurance Expenses from Total Assets Value	0.1%
Marketing Expenses from Total Revenues	2.00%

Table 25 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	100%
Income Tax after Deduction	0%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 26 Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premium	9.93%
Income Tax Rate	0.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 27 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premium	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	582,156
Loans Value	582,156
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	582,156
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	582,156
Equity Percentage	%50
Gross	
Project Value	1,164,312
Weighted Average Cost of Capital	%11.97

Capital Expenditures

The estimated cost of the project is 1,164,312 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 28 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	25	2,500	62,000

Total Cost

62,000

Table 29 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross / Cost
Administrative offices	150	150	22,500
Hangars, buildings	150	500	75,000
Refrigerated warehouses	150	1,000	150,000
Open spaces and garages	10	200	2,000
Cafeteria and workers' restrooms	100	150	15,000
Fence and gate	10	500	5,000
Other works	-	-	26,450
Total cost			295,950

Table 30 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Initial cleaning line	2	10000	20,000
Sorting and grading lines	4	50000	200,000
Refrigerated warehouses	8	15000	120,000
Washing and humidifying unit	1	5000	5,000
Sterilization unit	2	5000	10,000
Belt conveyer	4	5000	20,000
Packing and pressing machine	1	5000	5,000
Drying unit	2	10000	20,000
Tables and chairs	40	500	20,000
Control unit and laboratory	1	5000	5,000
Water cooling unit	1	3000	3,000
Packing and air-drawing machine	2	1000	2,000
Electronic balance	4	250	1,000
Forklift	2	1500	3,000
Field filling boxes, capacity of 5, 10, 20 kg	2.5	1000	2,500
Generator, air conditioners	1	10000	10,000
Other equipments	1	-	44,650

Total (JD) **491,150**

Table 31 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
Office furniture	2	2,000	4,000
Computer and its accessories	2	2,000	4,000
Kitchen and cafeteria equipment	2	2,000	4,000
Total			12,000

Table 32 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Medium vehicle	2	15,000	30,000
Cooled vehicle	1	40,000	40,000
Total cost	3		70,000

Table 33 Pre-Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,000
Transportation	1,000
Marketing	2,000
Training	1,000
Total cost	5,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 34 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	103,422
General and administrative expenses	3	%25	1,375
Marketing expenses	3	%25	2,068
Indirect Salaries	3	%25	15,000

Total	121,865
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Table 35 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	62,500	10%	68,750	%5.9
Construction works	295,950	10%	325,545	%28.0
Machinery & Equipment	491,150	10%	540,265	%46.4
Furniture	12,000	10%	13,200	%1.1
Vehicles	70,000	10%	77,000	%6.6
Pre-Operating Expenses	5,000	10%	5,500	%0.5
Working capital	121,865	10%	134,052	%11.5
Total (JD)	1,058,465		1,164,312	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,164,312) JD, where 50% of the project will be financed through loans (582,156) JD, and the remaining (50%) through (Equity) with (582,156) JD. The following table summarizes the general structure of the required financing:

Table 36 Sources of Funding

Funding Sources	Amount	Percentage
Equity	582,156	%50
Loans	582,156	%50
Total	1,164,312	%100
Use of Fund	Amount	Percentage
Capital Expenditures	1,024,760	%88.01
Pre-Operating Expenses	5,500	%0.47
Working Capital	134,052	%11.51
Total	1,164,312	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 37 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	169,680	329,664	329,664	329,664	329,664	329,664	329,664	329,664	329,664	329,664
Maintenance from Revenues	0	23,256	23,721	24,196	24,679	25,173	25,677	26,190	26,714	27,248
Disposables from Revenues	11,400	23,256	23,721	24,196	24,679	25,173	25,677	26,190	26,714	27,248
Direct Salaries	195,000	185,400	190,962	196,691	202,592	208,669	214,929	221,377	228,019	234,859
Others	37,608	56,158	56,807	57,475	58,161	58,868	59,595	60,342	61,111	61,902
Total	413,688	617,734	624,875	632,221	639,776	647,547	655,541	663,763	672,221	680,921

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 38 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Hospitality	500	510	520	531	541	552	563	574	586	598
Stationary and Printing	500	510	520	531	541	552	563	574	586	598
Consulting expenses	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Training	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Insurance	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Miscellaneous	500	510	520	531	541	552	563	574	586	598
Total	5,500	5,610	5,722	5,837	5,953	6,072	6,194	6,318	6,444	6,573

Marketing Expenses

Table 39 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	8,274	8,439	8,608	8,780	8,956	9,135	9,318	9,504	9,694	9,888

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 40 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	1	1	1	1	1	1	1	1	1	1
Technical food production worker	1	1	1	1	1	1	1	1	1	1
Accountant and warehouseguard	1	1	1	1	1	1	1	1	1	1
Quality auditor	1	1	1	1	1	1	1	1	1	1
Marketing officer	1	1	1	1	1	1	1	1	1	1
Security	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	6	6	6	6	6	6	6	6	6	6
direct employees										
Maintenance Technician	1	1	1	1	1	1	1	1	1	1
supervisor	1	1	1	1	1	1	1	1	1	1
Sorting,packaging workers	25	25	25	25	25	25	25	25	25	25
Driver	3	3	3	3	3	3	3	3	3	3

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Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cleaner	2	2	2	2	2	2	2	2	2	2
Total number of direct employees	32	32	32	32	32	32	32	32	32	32
Grand total	38	38	38	38	38	38	38	38	38	38

Table 41 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Technical food production worker	750	773	796	820	844	869	896	922	950	979
Accountant and warehouseguard	600	618	637	656	675	696	716	738	760	783
Quality auditor	500	515	530	546	563	580	597	615	633	652
Marketing officer	600	618	637	656	675	696	716	738	760	783
Security	300	309	318	328	338	348	358	369	380	391
direct employees										
Maintenance Technician	400	412	424	437	450	464	478	492	507	522
supervisor	400	412	424	437	450	464	478	492	507	522
Sorting,packaging workers	350	361	371	382	394	406	418	430	443	457
Driver	400	412	424	437	450	464	478	492	507	522
Cleaner	250	258	265	273	281	290	299	307	317	326

Table 42 Expected Annual Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Project Manager / Agricultural Engineer	16,000	16,480	16,974	17,484	18,008	18,548	19,105	19,678	20,268	20,876
Technical food production worker	12,000	12,360	12,731	13,113	13,506	13,911	14,329	14,758	15,201	15,657
Accountant and warehouseguard	9,600	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Quality auditor	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Marketing officer	9,600	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Security	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total salaries of indirect employees	60,000	61,800	63,654	65,564	67,531	69,556	71,643	73,792	76,006	78,286
direct employees										
Maintenance Technician	4,800	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
supervisor	4,800	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Sorting,packaging workers	105,000	144,200	148,526	152,982	157,571	162,298	167,167	172,182	177,348	182,668
Driver	14,400	19,776	20,369	20,980	21,610	22,258	22,926	23,614	24,322	25,052
Cleaner	6,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Total salaries of direct employees	135,000	185,400	190,962	196,691	202,592	208,669	214,929	221,377	228,019	234,859
Grand total	195,000	247,200	254,616	262,254	270,122	278,226	286,573	295,170	304,025	313,146

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 43 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	68,750	%0.0	%0.0
Construction Works	325,545	%5.0	%0.0
Machinaries	540,265	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	77,000	%10.0	%0.0
	1,024,760		

Table 44 Depreciations and Additions on Construction Works

Capex, Annual Additions and depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,024,760	1,036,225	1,047,940	1,059,909	1,072,140	1,084,638	1,097,410	1,110,463	1,123,804	1,137,439
Total Depreciation Expenses	79,324	80,470	81,642	82,839	84,062	85,312	86,589	87,894	89,228	90,592
Total Accumulated Depreciation	79,324	159,794	241,436	324,274	408,336	493,648	580,237	668,131	757,359	847,950
Total Additions	0	11,465	11,714	11,969	12,231	12,498	12,772	13,053	13,341	13,635
Total Net Book Values	945,436	876,431	806,504	735,635	663,804	590,990	517,174	442,333	366,445	289,489

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Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project

Table 45 Loan Details

Loan Value				582,156
Annual Interest Rate				9.50%
Loans Period				5
Grace Period				0
Loan Starts at year:				2018
Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				582,156
2019	151,615	55,305	96,310	485,846
2020	151,615	46,155	105,459	380,387
2021	151,615	36,137	115,478	264,909
2022	151,615	25,166	126,448	138,461
2023	151,615	13,154	138,461	0

Income Statement

Table 46 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	570,000	1,162,800	1,186,056	1,209,777	1,233,973	1,258,652	1,283,825	1,309,502	1,335,692	1,362,406
Gross Operating Revenues	570,000	1,162,800	1,186,056	1,209,777	1,233,973	1,258,652	1,283,825	1,309,502	1,335,692	1,362,406
Operating Expenses	(413,688)	(617,734)	(624,875)	(632,221)	(639,776)	(647,547)	(655,541)	(663,763)	(672,221)	(680,921)
Gross Operating Profit	156,312	545,066	561,181	577,557	594,197	611,105	628,284	645,738	663,470	681,484
<i>Gross Profit Percentage</i>	<i>%27</i>	<i>%47</i>	<i>%47</i>	<i>%48</i>	<i>%48</i>	<i>%49</i>	<i>%49</i>	<i>%49</i>	<i>%50</i>	<i>%50</i>
Salaries and Benefits (Indirect Staff)	(60,000)	(61,800)	(63,654)	(65,564)	(67,531)	(69,556)	(71,643)	(73,792)	(76,006)	(78,286)
General and Administraive	(5,500)	(5,610)	(5,722)	(5,837)	(5,953)	(6,072)	(6,194)	(6,318)	(6,444)	(6,573)
Markeing Expenses	(8,274)	(8,439)	(8,608)	(8,780)	(8,956)	(9,135)	(9,318)	(9,504)	(9,694)	(9,888)
Pre Operating Expenses	(5,500)									
Gross Indirect Expenses	(79,274)	(75,849)	(77,984)	(80,180)	(82,440)	(84,764)	(87,155)	(89,614)	(92,144)	(94,747)
Income before Interest,	77,038	469,217	483,197	497,376	511,757	526,341	541,129	556,124	571,326	586,737
Fixed Assets Depreciations	(79,324)	(80,470)	(81,642)	(82,839)	(84,062)	(85,312)	(86,589)	(87,894)	(89,228)	(90,592)
Income before Tax and Interests	(2,286)	388,747	401,555	414,537	427,695	441,029	454,541	468,230	482,098	496,145
Bank Interests	(55,305)	(46,155)	(36,137)	(25,166)	(13,154)	0	0	0	0	0
Income Before Tax	(57,590)	342,591	365,418	389,371	414,542	441,029	454,541	468,230	482,098	496,145
Income Tax	0	0	0	0	0	0	0	0	0	0
Net Profit	(57,590)	342,591	365,418	389,371	414,542	441,029	454,541	468,230	482,098	496,145

Dates Grading and Packing Center

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
<i>Net Profit Percentage</i>	<i>%10-</i>	<i>%29</i>	<i>%31</i>	<i>%32</i>	<i>%34</i>	<i>%35</i>	<i>%35</i>	<i>%36</i>	<i>%36</i>	<i>%36</i>
Compulsory Reserves	0	(34,259)	(36,542)	(38,937)	(41,454)	(44,103)	(45,454)	(46,823)	(48,210)	(49,615)
Retained Earnings	(57,590)	250,742	579,618	930,052	1,303,140	1,700,066	2,109,153	2,530,560	2,964,448	3,410,979

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 47 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities	(57,590)	342,591	365,418	389,371	414,542	441,029	454,541	468,230	482,098	496,145
Net Profit										
Bank Interests	55,305	46,155	36,137	25,166	13,154	0	0	0	0	0
Depreciation	79,324	80,470	81,642	82,839	84,062	85,312	86,589	87,894	89,228	90,592
Total Operating Cashflows before Additions to Working Capital	77,038	469,217	483,197	497,376	511,757	526,341	541,129	556,124	571,326	586,737
Inventory (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Receivables (Increase/Decrease)	(47,500)	(49,400)	(1,938)	(1,977)	(2,016)	(2,057)	(2,098)	(2,140)	(2,183)	(2,226)
Accounts Payables (Increase/Decrease)	34,474	17,004	595	612	630	648	666	685	705	725

Dates Grading and Packing Center

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Working Capital (Increase/Decrease)	(13,026)	(32,396)	(1,343)	(1,365)	(1,387)	(1,409)	(1,432)	(1,455)	(1,478)	(1,501)
Net Cashflows from Operating Activities	64,012	436,821	481,854	496,011	510,370	524,932	539,698	554,670	569,848	585,236
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(1,024,760)	(11,465)	(11,714)	(11,969)	(12,231)	(12,498)	(12,772)	(13,053)	(13,341)	(13,635)
Net Cashflows from Investments Activities	(1,024,760)	(11,465)	(11,714)	(11,969)	(12,231)	(12,498)	(12,772)	(13,053)	(13,341)	(13,635)
Cashflows from Financing Activities										
Capital	582,156									
Loan Amortization	(96,310)	(105,459)	(115,478)	(126,448)	(138,461)	0	0	0	0	0
Bank Interest Rate	(55,305)	(46,155)	(36,137)	(25,166)	(13,154)	0	0	0	0	0
Loans	582,156	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,012,697	(151,615)	(151,615)	(151,615)	(151,615)	0	0	0	0	0
Net (Increase/Decrease) in Cash	51,950	273,741	318,525	332,427	346,525	512,434	526,926	541,617	556,508	571,600
Cashflows at the Beginning of Period	0	51,950	325,691	644,215	976,643	1,323,168	1,835,602	2,362,527	2,904,144	3,460,651
Cashflows at the End of Period	51,950	325,691	644,215	976,643	1,323,168	1,835,602	2,362,527	2,904,144	3,460,651	4,032,252

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 48 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	51,950	325,691	644,215	976,643	1,323,168	1,835,602	2,362,527	2,904,144	3,460,651	4,032,252
Inventory	0	0	0	0	0	0	0	0	0	0
Accounts Receivable	47,500	96,900	98,838	100,815	102,831	104,888	106,985	109,125	111,308	113,534
Total current Assets	99,450	422,591	743,053	1,077,458	1,425,999	1,940,489	2,469,513	3,013,269	3,571,959	4,145,786
Non Current Assets										
Fixed Assets (net)	945,436	876,431	806,504	735,635	663,804	590,990	517,174	442,333	366,445	289,489
Total Non Current Assets	945,436	876,431	806,504	735,635	663,804	590,990	517,174	442,333	366,445	289,489
Total Assets	1,044,886	1,299,022	1,549,557	1,813,092	2,089,803	2,531,480	2,986,687	3,455,602	3,938,404	4,435,275
Liabilities										
Current Liabilities										
Paybles	34,474	51,478	52,073	52,685	53,315	53,962	54,628	55,314	56,018	56,743

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Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Remaining amount of Loan	105,459	115,478	126,448	138,461	0	0	0	0	0	0
Total current Liabilities	139,933	166,956	178,521	191,146	53,315	53,962	54,628	55,314	56,018	56,743
Non Current Liabilities										
Long Terms Loans	380,387	264,909	138,461	0	0	0	0	0	0	0
Total Long Term Liabilities	380,387	264,909	138,461	0	0	0	0	0	0	0
Total Liabilities	520,320	431,865	316,982	191,146	53,315	53,962	54,628	55,314	56,018	56,743
Owners Equity										
Shareholders Contributions	582,156	582,156	582,156	582,156	582,156	582,156	582,156	582,156	582,156	582,156
Statutory Reseve	0	34,259	70,801	109,738	151,192	195,295	240,749	287,572	335,782	385,397
Retained Profits	(57,590)	250,742	579,618	930,052	1,303,140	1,700,066	2,109,153	2,530,560	2,964,448	3,410,979
Total Equity	524,566	867,157	1,232,575	1,621,946	2,036,488	2,477,517	2,932,058	3,400,288	3,882,386	4,378,531
Total Liabilities and Equity	1,044,886	1,299,022	1,549,557	1,813,092	2,089,803	2,531,480	2,986,687	3,455,602	3,938,404	4,435,275

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

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Table 49 Free Net Cashflows

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,164,312)	69,512	425,356	470,139	484,042	498,140	512,434	526,926	541,617	556,508	571,600	7,463,459
Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash Flows	(1,164,312)	62,084	339,303	334,950	308,002	283,100	260,102	238,876	219,297	201,248	184,616	2,410,556

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 50 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(1,164,312)	69,512	425,356	470,139	484,042	498,140	512,434	526,926	541,617	556,508	571,600
Project Value Returned	1,164,312	1,094,800	669,444	199,305	(284,737)	(782,877)	(1,295,311)	(1,822,236)	(2,363,853)	(2,920,361)	(3,491,961)
Payback Period (Year)	3	1	1	1	0	0	0	0	0	0	0

Table 51 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.97
Net Present Value for Cashflows	3,677,823
Payback Period	3
Internal Rate of Return	%37.15

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 52 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC	Sensitivity Analysis
Original Scenario	37.15%	3	3,678	11.97%
Revenues declined by 10%	30.42%	4	2,523	11.97%
Operating Expenses Increased by 10%	33.04%	3	3,015	11.97%
Total cost Increased by 10%	30.97%	4	2,897	11.97%
Total cost Increased and revenues decreased by 10%	24.30%	5	1,748	11.97%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,677,823 Jordan Dinars considering that the project provides 38 Job Opportunities for the governorate residents.

